

William D. Zhang

<https://willyum00.github.io/>

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Education	University of California San Diego , San Diego, CA, USA Master of Science (MS) Electrical Engineering Specialization: Intelligent Systems, Robotics, and Controls Bachelor of Science (BS) Electrical Engineering	Sep 2024 - Present Sep 2020 - June 2024
Experience	Visual Intelligence Studio with Dr. Yang Cai Worked on <i>EXR Telemetry Interface for Real-Time Operation</i> funded by Wellcome Leap SAVE . - Built a 3D vision based program for surgeons to train in laparoscopic surgery. - Wrote tracking algorithm using stereo depth cameras for precise spatial feedback of surgical tools and organs in 3D space. - Created interactive holographic visuals of patient organs with hand tracking. - Presented at 5x5 Public Safety conference and meetings with PI's at UCSD and Stanford.	Feb 2023 - June 2024
Skills	Languages: Python, Java, C, C++, Bash, MATLAB, $\text{\LaTeX}$ Software: Unity, SystemVerilog, Verilog RTL, LTspice Libraries: OpenCV, Jax, Intel RealSense SDK, PyLSL, EEGLab, JUnit Graduate Coursework: Sensing & Estimation in Robotics, Digital Image Processing	
Selected Projects	EMG Mouse: AeroMus — NTX Competition Project with Triton Neurotech - Built a wearable computer mouse that enables people with motor disabilities to use a computer via muscle contractions. - Designed a multichannel stainless steel electrode for improved resolution and reduced noise. - Built circuit that connects electrodes to bio-sensor with minimal wiring and low footprint. - Won second place across twenty teams internationally. Low Cost Neural Interface - Built a custom, low cost, 2 channel A/D converter for detecting EMG signals. - Wrote program to send and receive live sensor data over WiFi. Kalman Filter Tracking - Improved Kalman Filter performance with occluded objects by implementing shadow tracking. - Tested and showed increased tracking performance in different lighting and environments.	
Leadership	Triton NeuroTech Vice President - Facilitated relationships between club, ECE department, companies and other organizations. - Secured thousands of dollars in funding through fundraising and grants. - Grew club membership from 10 to over 50 active members in less than two years. Poster Presentation: <i>Your Brain on DALL-E</i>, presented at California Neurotech Conference.	May 2022 - Feb 2024
Community Involvement	HKN, <i>ECE Outreach Mentor</i> Taught basic electrical engineering concepts at low-income high schools across San Diego. HARD Hack, <i>Hackathon Volunteer</i> Triton Neurotech, <i>Community and Design Chair</i> UCSD Envision Maker Space, <i>Student Mentor</i> Alpha Math and Science Institute, <i>Calculus and Physics Tutor</i>	Jan 2024 Apr 2023 Oct 2020 - May 2022 Oct 2021 - June 2022 June 2019 - May 2020